

Oracle EPM Predictive Cash Forecasting

Opening / Closing Cash Position - Advanced Lab Guide

Product Design Documentation • Implementation Guide • Training Material • Technical Reference • Troubleshooting Handbook

Designed for advanced-stage Oracle EPM, treasury, FP&A, and cash-management learners who need both functional understanding and implementation-level control of cash-position forecasting.

Core equation Closing Cash Position = Opening Cash Position + Total Cash Inflows - Total Cash Outflows. The next period normally begins from the prior period closing position, subject to validated bank-balance and forecast-range logic.

Predictive Cash Forecasting / Cash Position

Opening Cash

\$42.6M

Net Cash Flow

-\$5.8M

Closing Cash

\$36.8M

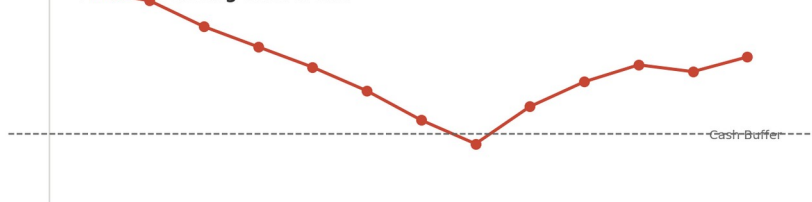
Cash Buffer

\$30.0M

Cash Over Buffer

\$6.8M

18-Week Closing Cash Trend



Exceptions

W08	Below buffer	Critical
W09	Large AR shift	Warning
W12	Bank balance state	Warning

Management Insight

Week 8 closing cash breaches the 30M buffer. Stress case recommends a 5M revolver draw or accelerated collections.

Guide Focus	What the Student Will Build
Cash Position Model	Opening, movement, closing, buffer and headroom logic
Integration	Bank / ERP balance and transaction flow with validation
Forecast Operations	Daily/weekly/periodic position processing and roll-forward
Controls	Reconciliation, stale-balance checks, exceptions, workflow
Decisioning	Funding, cash transfer, debt repayment and investment triggers

1. Solution Context and Learning Objectives

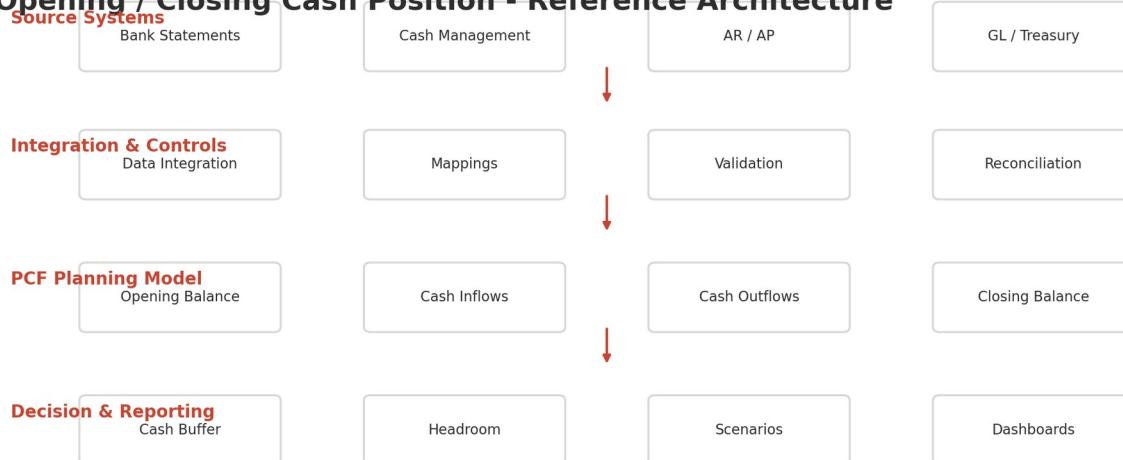
Predictive Cash Forecasting is a Planning application type for short- and medium-term cash forecasting. In this lab, the cash-position layer is treated as the control point that connects all Operating, Investing and Financing cash flows to treasury decisions.

Oracle alignment Current Oracle documentation describes cash line items that include Opening Cash Balance and Closing Cash Balance, supports daily/weekly/monthly rolling forecasts, and provides cash-buffer / cash-over-buffer analysis. Oracle Cloud ERP integration can seed AR, AP and Cash Management data and support drill-through.

Advanced learning outcomes

- Design a cash-position hierarchy that reconciles legal-entity, currency and bank-account views.
- Configure opening balance sourcing and carry-forward behavior without double-counting transactions.
- Calculate and validate closing position, cash buffer and cash over buffer.
- Build Base, Stress and Management Action scenarios.
- Diagnose stale bank balances, missing opening balances, FX mismatches and roll-forward breaks.
- Operate the process using jobs, approval workflow, audit and production runbooks.

Opening / Closing Cash Position - Reference Architecture

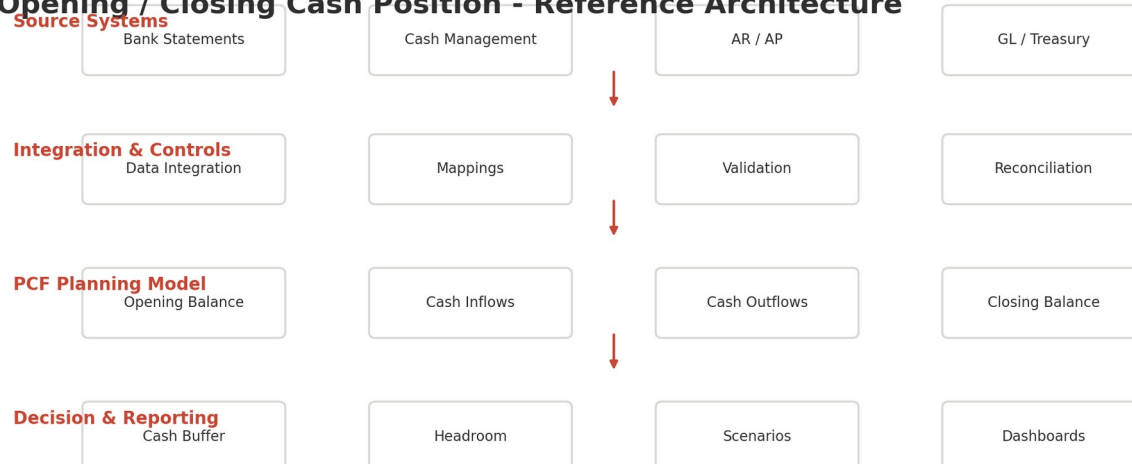


Governance: Security • Workflow • Audit • Job Monitoring • Data Lineage

2. Functional Architecture

A professional cash-position design separates source balances, transactional cash-flow forecasts, calculation logic, controls and decision outputs. The opening position must be traceable to an approved source and the closing position must reconcile to the line-item movement.

Opening / Closing Cash Position - Reference Architecture



Governance: Security • Workflow • Audit • Job Monitoring • Data Lineage

Layer	Responsibilities	Key Control
Bank / ERP Sources	Opening bank balances, receipts, payments, treasury transactions	As-of date and balance type
Integration	Extract, periodize, map, load, drill-through	Duplicate and stale data checks
PCF Planning Model	Line items, entities, scenarios, versions, currencies	Balanced cash hierarchy
Rules	Process Forecast, aggregation, carry-forward, FX	Opening-to-closing continuity
Dashboard / Actions	Headroom, risk, liquidity actions, variance	Approval and audit trail

3. Cash Position Data Model

Recommended line-item structure

Parent / Line Item	Type	Sign / Calculation	Purpose
Opening Cash Balance	Balance	Input / sourced	Starting cash available
Operating Cash Flows	Flow	Sum of operating inflows/outflows	Working capital movement
Investing Cash Flows	Flow	Sum of investing activity	Capex, asset sales, investments
Financing Cash Flows	Flow	Sum of financing activity	Debt, equity, dividends, leases
Net Cash Flow	Calculated	Operating + Investing + Financing	Period movement
Closing Cash Balance	Calculated	Opening + Net Cash Flow	Ending cash position
Cash Buffer	Assumption	Input	Minimum target liquidity
Cash Over Buffer	Calculated	Closing - Cash Buffer	Headroom / risk measure

Dimensional design

- Entity: Group > Region > Legal Entity; consider bank-account reporting as source detail or reporting attribute.
- Period: Daily, weekly or monthly depending forecast range.
- Scenario / Version: Actual, Forecast, Base, Stress, Management Action.
- Currency: Local and reporting currency with controlled FX treatment.
- Cash Line Item: Opening, detailed flows, Net Cash Flow, Closing, Cash Buffer and Cash Over Buffer.
- Data Source: Bank, ERP actual, ERP open item, Forecast Method, Manual Adjustment.

4. Opening Cash Position - Source and Control Design

Opening cash is not merely a user-entered number. It is a controlled starting balance that determines the reliability of every downstream forecast period.

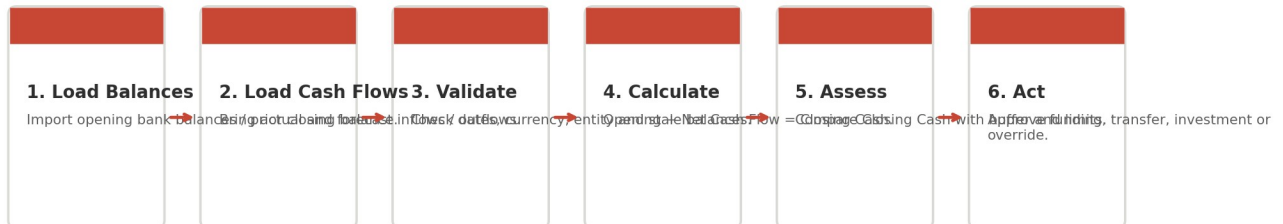
Source Option	When to Use	Control Consideration
Bank statement closing available balance	Liquidity-focused treasury view	Most recent statement date and stale-balance threshold
Bank statement closing booked balance	Ledger-oriented bank position	Timing differences versus available cash
Prior PCF closing balance	Rolling forecast continuity	Must reconcile to prior approved forecast
GL cash balance	Fallback / reconciliation view	May not represent immediately available liquidity
Manual opening adjustment	Exceptional transition / correction	Require comment, attachment and approval

Important design rule When a source bank balance is older than the forecast start date, explicitly account for intervening cash flows. Avoid overlapping overdue transactions with prior-day cash-flow adjustments, which can double-count movement.

5. Closing Cash Position - Calculation Logic

Cash Position Processing Flow

Direct-method cash position logic with governance checkpoints



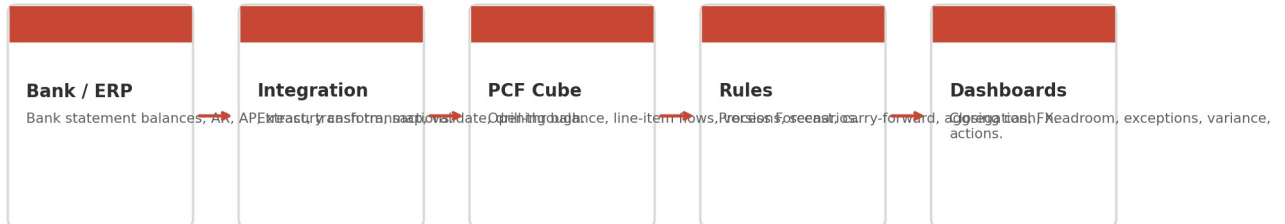
Calculation sequence

1. Resolve opening balance as of the forecast start date.
2. Aggregate operating, investing and financing cash inflows / outflows for the period.
3. Calculate Net Cash Flow.
4. Calculate Closing Cash = Opening Cash + Net Cash Flow.
5. Compare Closing Cash with Cash Buffer and internal liquidity limits.
6. Carry the approved closing position into the next forecast period, unless the next period opening is refreshed from an authoritative actual bank balance.

Reconciliation test Opening Cash + Net Cash Flow - Closing Cash must equal zero for every entity / period / scenario combination unless an approved bridge item explains the difference.

6. Data Integration and Lineage

Data Flow - Source to Closing Position



Data Set	Typical Source	Frequency	Drill / Evidence
Opening bank balance	Cash Management / bank statement	Daily or forecast start	Bank account and statement date
Customer receipts	Receivables	Daily	Customer / receipt / invoice
Supplier payments	Payables	Daily	Supplier / payment / invoice
Cash transactions	Cash Management	Daily	Transfer / miscellaneous transaction
Forecast assumptions	EPM / Treasury input	As needed	Driver, method, owner, approval
Manual adjustment	EPM form	Exception only	Comment, attachment, approver

7. Oracle-Inspired Configuration Mockup

Application > Configure > Cash Position Setup

Forecast Frequency
Weekly

Forecast Horizon
13 Weeks

Opening Balance Source
Cash Management

Closing Balance Code
Closing Available

Cash Buffer
\$30,000,000

Reporting Currency
USD

Scenario
Base

Stale Balance Threshold
2 Days

Save & Process

Implementation note Use Oracle-inspired navigation and terminology for training, but do not reproduce proprietary Oracle screens pixel-for-pixel. Build your own forms, cards and dashboards around the functional concepts.

8. Advanced Lab Scenario - 13-Week Cash Position

BISP Global Services has a \$42.6M opening cash position and a policy cash buffer of \$30.0M. Treasury must forecast 13 weeks, identify the first liquidity breach, and recommend an action.

Week	Opening	Operating	Investing	Financing	Net Flow	Closing	Buffer Gap
W01	42.6	1.4	-2.0	-0.2	-0.8	41.8	11.8
W02	41.8	0.9	-2.8	-0.4	-2.3	39.5	9.5
W03	39.5	1.1	-2.4	-0.5	-1.8	37.7	7.7
W04	37.7	0.7	-1.9	-0.6	-1.8	35.9	5.9
W05	35.9	0.6	-2.1	-0.6	-2.1	33.8	3.8
W06	33.8	0.3	-2.3	-0.6	-2.6	31.2	1.2
W07	31.2	-0.1	-1.5	-0.5	-2.1	29.1	-0.9
W08	29.1	1.8	0.9	0.6	3.3	32.4	2.4
W09	32.4	1.5	0.9	-0.2	2.2	34.6	4.6
W10	34.6	1.3	0.8	-0.6	1.5	36.1	6.1
W11	36.1	0.5	-0.8	-0.3	-0.6	35.5	5.5
W12	35.5	1.6	0.3	-0.6	1.3	36.8	6.8

Student task Validate each period roll-forward, identify the first buffer breach, and determine whether the issue is timing, structural cash burn, stale opening balance, or a forecast-method problem.

9. Lab Calculation Walkthrough

Step 1 - W01 closing cash

Net Cash Flow = $1.4 - 2.0 - 0.2 = -0.8M$. Closing Cash = $42.6 + (-0.8) = 41.8M$. Cash Over Buffer = $41.8 - 30.0 = 11.8M$.

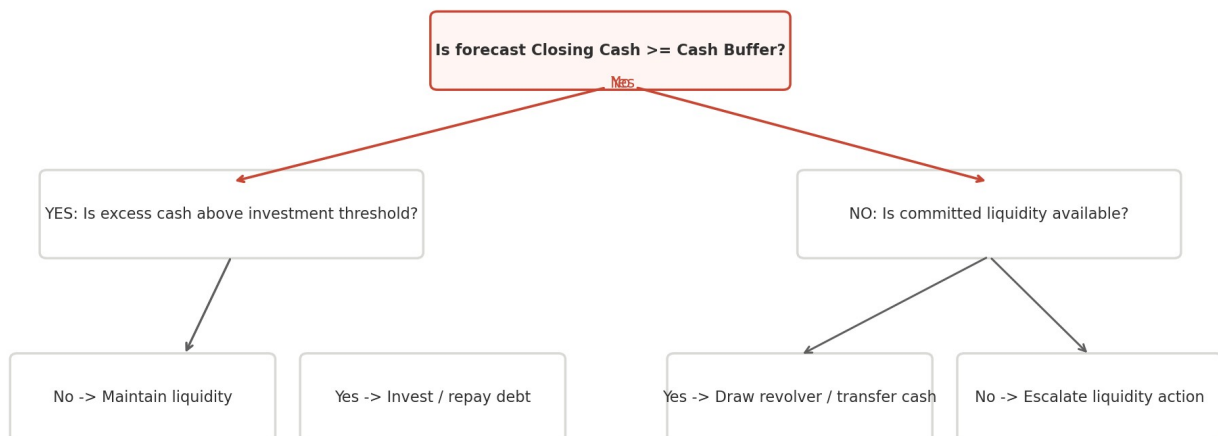
Step 2 - Roll to W02

W02 Opening Cash should equal W01 approved Closing Cash, 41.8M. Any difference becomes a continuity exception unless it is explained by a refreshed actual bank balance.

Step 3 - Identify breach

W07 Closing Cash falls to 29.1M, which is 0.9M below the \$30.0M buffer. Treasury should examine whether the breach is temporary and whether W08 inflows are sufficiently certain.

Closing Cash Position Decision Tree



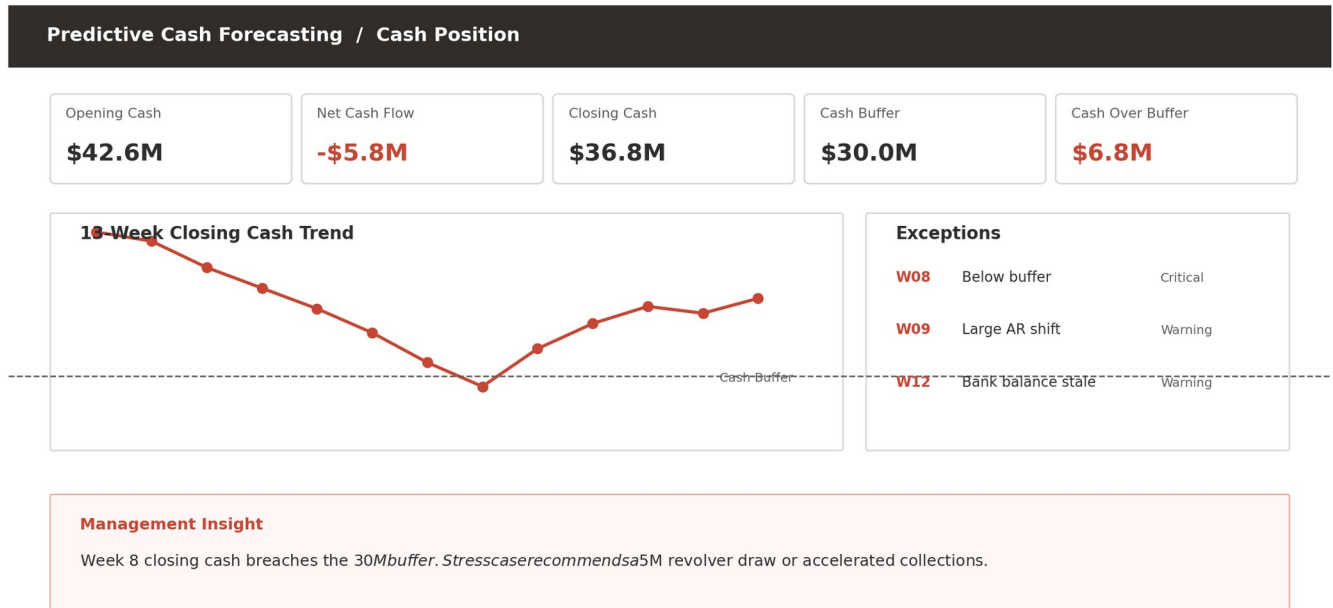
Every action should be scenario-tested, approved, and traceable to source data.

10. Scenario Stress and Management Action

Scenario	Key Changes	W07 Closing	Action
Base	Current receipt/payment assumptions	\$29.1M	Prepare contingency funding
Stress	Customer receipts -15%; Capex unchanged	\$25.4M	Draw revolver before W06 close
Management Action	Accelerate \$3M collections + defer \$2M discretionary Capex	\$34.1M	Avoid borrowing; monitor execution
Severe Stress	Receipts -25%; unexpected tax outflow	\$21.8M	Escalate to CFO / liquidity committee

Advanced point Scenario actions must be executable. A management scenario that assumes accelerated collections or delayed spend should have an owner, due date, evidence and approval status.

11. Cash Position Dashboard and Management Cockpit



Recommended KPIs

- Opening cash, Net Cash Flow, Closing cash, Cash Buffer and Cash Over Buffer.
- Minimum closing cash in horizon and first breach date.
- Available revolver / committed liquidity and utilization.
- Forecast variance, forecast accuracy and largest line-item drivers.
- Bank balance freshness and unreconciled cash-position exceptions.

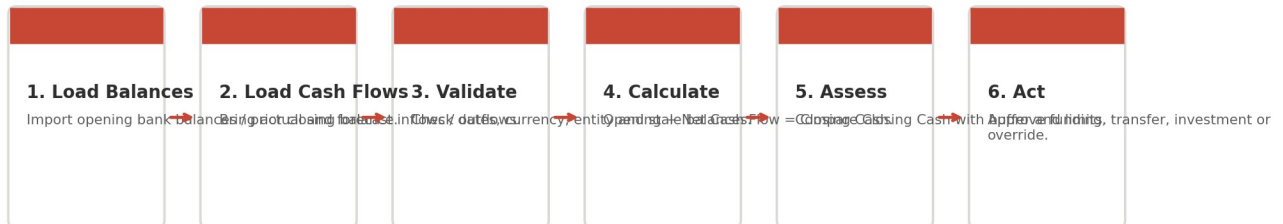
12. Reconciliation and Control Framework

Control	Formula / Test	Severity	Owner
Roll-forward continuity	Prior Closing = Current Opening	Error	Cash Manager
Cash equation	Opening + Net Flow - Closing = 0	Error	Controller
Bank freshness	Balance date within threshold	Error / Warning	Treasury Ops
Currency completeness	Every entity has valid currency / FX	Error	EPM Admin
Buffer coverage	Closing >= Cash Buffer	Warning / Critical	Treasury
Manual override	Comment + attachment + approver	Error	Cash Manager
Duplicate transactions	Unique source keys / no overlap	Error	Integration Owner

13. Workflow, Approval and Audit

Cash Position Processing Flow

Direct-method cash position logic with governance checkpoints



Role	Responsibilities	Typical Rights
Cash Preparer	Review sources, enter assumptions, explain exceptions	Input / submit
Cash Manager	Approve forecast, manage buffer / scenarios	Approve / override
Controller	Reconcile to GL / ERP actuals	Review / certify
Treasury Head	Approve liquidity action	Scenario / decision approval
Administrator	Metadata, rules, integrations, security	Administrative

Audit design Log who changed opening balance, forecast method, cash buffer, scenario assumptions or manual adjustments; capture old value, new value, timestamp, source, comment and approval status.

14. Exception Management

Exception	Example	Root Cause	Resolution
Missing opening balance	Entity US01 shows blank opening	Bank account not mapped	Map account and reload balance
Stale balance	Last bank balance is 5 days old	Statement feed failure	Load new balance or approved bridge cash flows
Roll-forward break	W05 opening != W04 closing	Manual overwrite / refreshed actual	Reconcile source date and document bridge
Negative closing cash	W07 = -\$1.2M	Large payment / missing facility draw	Scenario and funding action
FX imbalance	Group closing differs from entity roll-up	Wrong rate / currency mapping	Validate currency and translation
Duplicate movement	Receipts appear twice	Overdue + prior-day overlap	Fix extract / transformation logic

15. Log and Error Examples

Timestamp	Job / Rule	Status	Message
07:00:12	Load Bank Balances	ERROR	US01 opening balance missing for Bank_102
07:03:40	Load AR/AP	SUCCESS	84,217 transactions loaded
07:05:11	Validate Cash Position	WARNING	DE01 bank balance age = 4 days; threshold = 2
07:06:02	Process Forecast	ERROR	Roll-forward variance W07 = 0.9M
07:08:26	Cash Buffer Check	WARNING	US01 W07 closing 29.1M < buffer 30.0M

Troubleshooting sequence Start with source freshness and mapping, then confirm POV / scenario / version, then validate opening-to-closing arithmetic, then inspect forecast methods and manual adjustments. Re-run only after the root cause is understood.

16. Implementation Guide - Build Sequence

1. Confirm treasury policy: balance type, cash buffer, reporting currency and required forecast frequency.
2. Design entity and cash-line-item hierarchy.
3. Configure source integration for bank balances, Receivables, Payables and Cash Management.
4. Map bank accounts / legal entities and validate data freshness.
5. Configure opening / closing balance logic and roll-forward controls.
6. Set forecast methods for operating, investing and financing line items.
7. Build Base, Stress and Management Action scenarios.
8. Create cash-position dashboard and exception workbench.
9. Configure security, workflow, audit and job schedules.
10. Execute parallel run against existing treasury forecast and reconcile differences.
11. Document production runbook, support ownership and recovery procedures.

17. Technical Reference - Calculation and Rule Design

Object	Design Guidance
Opening Cash Balance	Sourced from authoritative balance or prior closing; protect from uncontrolled input
Net Cash Flow	Sum signed cash-flow line items for period / entity / scenario
Closing Cash Balance	Opening + Net Cash Flow
Cash Buffer	Policy assumption; may vary by entity and period
Cash Over Buffer	Closing - Cash Buffer
Roll-forward	Prior approved closing -> next opening unless refreshed by actual balance
Aggregation	Aggregate legal entity cash position with currency / ownership rules as applicable
Validation	Zero continuity variance and no missing source balance
Override	Separate adjustment member / data source preferred for auditability

18. Automation and Operational Runbook

Sequence	Automation Step	Failure Handling
1	Extract / load bank balances	Stop if no current balance for critical entities
2	Load AR/AP/Cash Management transactions	Reject duplicates; retain source keys
3	Run mappings / validations	Block process on Error severity
4	Process Forecast	Capture job ID, elapsed time and rule status
5	Run reconciliation / buffer checks	Generate exception list
6	Refresh dashboards / notify users	Send only after successful controls
7	Submit / approve forecast	Lock approved scenario or version
8	Archive evidence	Retain logs, source extracts and approval record

Recovery principle Do not simply rerun a failed process repeatedly. Classify the failure as source, mapping, rule, performance, security or data-quality; correct it; then restart from the earliest affected step.

19. Product Design Requirements

Requirement	Acceptance Criteria
Single cash-position truth	Every dashboard ties to the same opening / closing line-item hierarchy
Traceable opening balance	Source, account, date and balance type are visible
No silent continuity breaks	Opening / closing mismatch generates exception
Actionable liquidity risk	First buffer breach and amount visible without manual calculation
Scenario comparison	Base vs Stress vs Action available in same POV
Drill-through	Users can inspect source balances / transactions where integration supports it
Controlled override	All manual changes require reason and approval
Production observability	Jobs, errors, duration and last successful refresh are reportable

20. Advanced Student Exercises

- Change the bank balance source from Closing Available to Closing Booked and explain the liquidity impact.
- Introduce a 3-day stale bank balance and design a bridge using intervening cash flows without double-counting.
- Create a Stress scenario with a 20% delay in customer receipts and identify the first buffer breach.
- Add a \$5M revolver draw in W07 and verify closing-cash roll-forward through W13.
- Create an FX mismatch for a EUR entity and diagnose the group cash-position difference.
- Design a manual adjustment control that requires comment, attachment and manager approval.
- Build a variance report comparing prior forecast closing cash to actual bank closing balance.
- Define three operational alerts: stale balance, cash below buffer, and roll-forward difference.

21. Troubleshooting Handbook - Symptom to Resolution

Symptom	Diagnostic Questions	Likely Fix
Opening cash blank	Is source loaded? Entity/bank mapped? Correct POV?	Fix mapping / load / source date
Closing cash too high	Duplicate receipts? Wrong signs? Missing outflows?	Trace line items and source keys
Closing cash too low	Missing receipts? Large manual adjustment? FX issue?	Compare source to loaded line items
Opening does not equal prior closing	Actual refresh? Manual overwrite? Scenario mismatch?	Reconcile bridge and protect input
Buffer dashboard unchanged	Was buffer entered for correct entity/year and forecast processed?	Correct POV and rerun forecast
Group position differs from entity total	Currency translation or hierarchy issue?	Review rates / entity hierarchy
Forecast rule fails	Invalid member / missing dependency / data issue?	Review job log and rule context

22. Reference Notes and Release-Aware Design

The guide is designed against current Oracle EPM Predictive Cash Forecasting concepts available in 2026. Always validate your pod release and Oracle documentation before production implementation because capabilities evolve.

Oracle Reference	Relevant Point
Working with Predictive Cash Forecasting (Apr 2026)	Cash line items include Opening Cash Balance and Closing Cash Balance; Operating/Investing/Financing structure.
Administering Predictive Cash Forecasting (Apr 2026)	ERP extraction options include bank closing balance type such as Closing Available or Closing Booked.
Predictive Cash Forecasting - ERP Integration (Apr 2026)	Ready-to-use integration with Fusion AR, AP and Cash Management plus drill-through.
Cash Buffer enhancement (Jun 2025)	Cash Buffer input and Cash Over Buffer = Closing Balance - Cash Buffer.
PCF overview	Daily, weekly and monthly rolling forecasts; direct cash flow method; scenario and predictive methods.

Training disclaimer All figures, data, mockups and examples in this guide are instructional and fictional. Oracle-inspired UI means conceptually aligned enterprise navigation and controls, not a reproduction of Oracle proprietary screen design.